



Report

Nara Avila Moraes

Physics Institute
IFUSP

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Summary

① Tasks Status

② Results

③ Selected Article

General Tasks Status

Task	Status	
Correlation Inference Study - Classical Approach		
Correlation Inference Study - Quantum Learning Approach		
2026-11 Paper		
2026-12 LASNPA Presentation		

Correlated Uncertainty Study Tasks Status

Task	Status	Link
Alice ins1222333 Studies		
↪ Uncertainty correlation effect	✓	Plots
RHIC Au-Au/d-Au Studies JETSCAPE		
↪ Uncertainty correlation effect	✓	Plots
Atlas ins1759506 studies		
↪ Uncertainty correlation inference	🔄	Notebook
Alice PbPb2760 Scaled Spectra studies		
↪ Uncertainty correlation inference	🔄	Notebook
Synthetic data studies		
↪ Uncertainty correlation inference	○	

Atlas ins1759506 studies

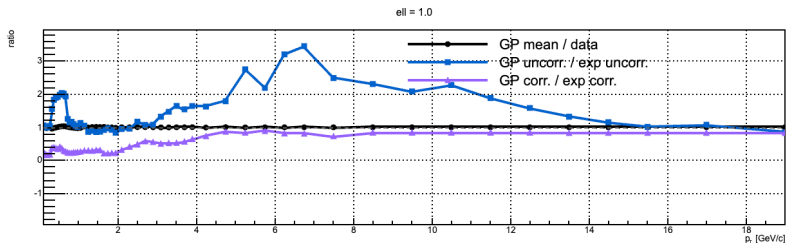


Figure: GP fit: fixed ell = 1.0

Atlas ins1759506 studies

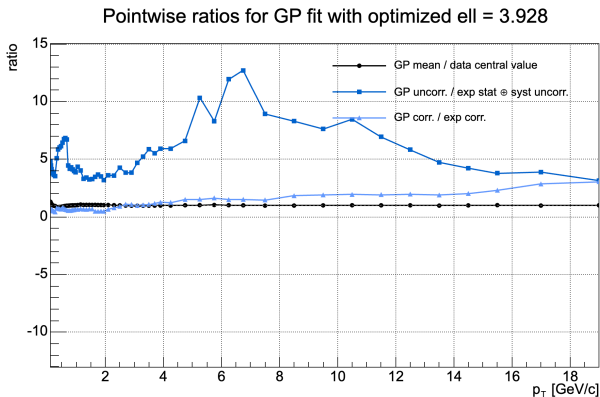


Figure: GP fit: optimized ell

Atlas ins1759506 studies

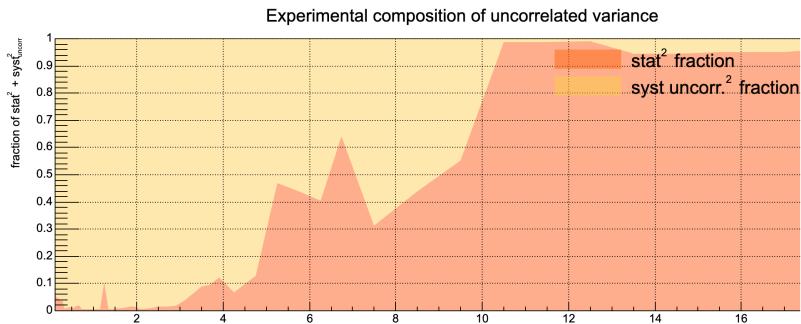


Figure: Ratio between white noise and systematic uncorrelated noise

Alice PbPb2760 Scaled Spectra studies

ALICE Pb-Pb 2.76 TeV Grad, 0-5%

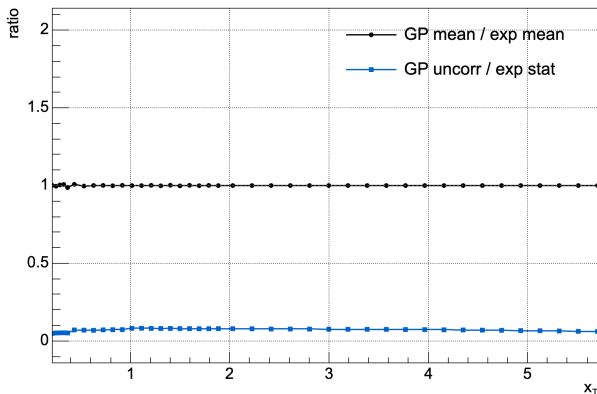


Figure: GP fit: fixed ell = 0.2

Alice PbPb2760 Scaled Spectra studies

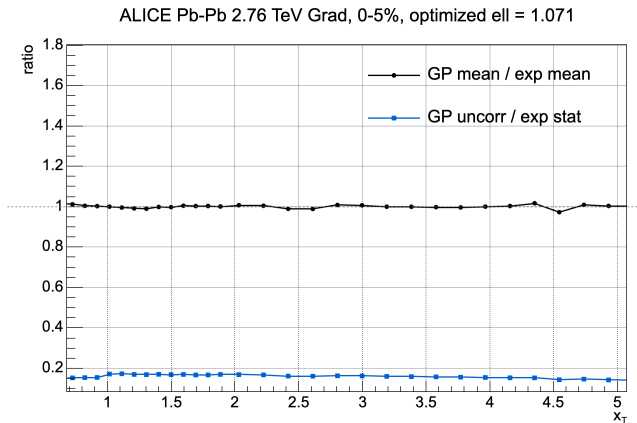


Figure: GP fit: optimized ell

Selected Article

NNStar: An end-to-end AI agent for nuclear matter and neutron star physics

Yao Ma,^{1,✉} Yong-Liang Ma,^{2,1,†} and Jia-Ying Xiong^{2,1,‡}

¹*School of Frontier Sciences, Nanjing University, Suzhou 215163, China*

²*School of Physics, Nanjing University, Nanjing 210093, China*

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Constraining the equation of state of dense matter requires confronting effective models with massive data that spans many orders of magnitude in scale, from sub-saturation nuclear matter properties to the masses, radii, and tidal deformabilities of neutron stars. Exploring the high-dimensional coupling space of such a model and fine tuning it against all of these constraints is a labor- and time-intensive task. We present NNSTAR, an end-to-end artificial-intelligence agent that automates this workflow. Rather than a bespoke application, NNSTAR is delivered as a portable *skill* for an open large-language-model (LLM) agent platform—a self-describing module that pairs worked usage conventions with symbolic and numerical physics engines that (i) build a relativistic mean-field model directly from a Lagrangian, (ii) solve the mean-field equations of motion and evaluate the saturation properties, (iii) construct the β -equilibrium equation of state, splice it to a crust, and integrate the Tolman–Oppenheimer–Volkoff equations, and (iv) score the resulting predictions through a Bayesian joint analysis against nuclear matter and astrophysical observations. The agent can read a model, fit its parameters, and report the full set of nuclear matter and neutron star observables without human intervention. NNSTAR therefore provides a new, AI-driven framework for analyzing nuclear matter and neutron-star observations.

Figure: AI Agent to research nuclear matter physics